JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

IIT JAM 2023 Official Paper

Category: CBT Era (2015–2026) • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
60	100 M	180 Min	MCQ • MSQ • NAT (60 Ques

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	30	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	10	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	20	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2023 • Q1 • ID: JAM_2023_Q1) MCQ +1 M Neg: −0.33

Q. 1 Let G be a finite group. Then G is necessarily a cyclic group if the order of G is

(A) 4

(B) 7

(C) 6

(D) 10

Question 2 (JAM 2023 • Q2 • ID: JAM_2023_Q2) MCQ +1 M Neg: −0.33

Q. 2 Let $\mathbf{v}_1, \ldots, \mathbf{v}_9$ be the column vectors of a non-zero 9×9 real matrix A. Let $a_1, \ldots, a_9 \in \mathbb{R}$, not all zero, be such that $\sum_{i=1}^{9} a_i \mathbf{v}_i = \mathbf{0}$. Then the system $A\mathbf{x} = \sum_{i=1}^{9} \mathbf{v}_i$ has

(A) no solution

(B) a unique solution

(C) more than one but only finitely many solutions

(D) infinitely many solutions

Question 3 (JAM 2023 • Q3 • ID: JAM_2023_Q3) MCQ +1 M Neg: −0.33

Q. 3 Which of the following is a subspace of the real vector space $\mathbb{R}^3$?

(A) $\{(x, y, z) \in \mathbb{R}^3 : (y + z)^2 + (2x - 3y)^2 = 0\}$

(B) $\{(x, y, z) \in \mathbb{R}^3 : y \in \mathbb{Q}\}$

(C) $\{(x, y, z) \in \mathbb{R}^3 : yz = 0\}$

(D) $\{(x, y, z) \in \mathbb{R}^3 : x + 2y - 3z + 1 = 0\}$

Question 4 (JAM 2023 • Q4 • ID: JAM_2023_Q4) MCQ +1 M Neg: −0.33

Q. 4 Consider the initial value problem

$$\frac{dy}{dx} + \alpha y = 0,$$
$$y(0) = 1,$$

where $\alpha \in \mathbb{R}$. Then

(A) there is an α such that $y(1) = 0$

(B) there is a unique α such that $\lim_{x \to \infty} y(x) = 0$

(C) there is NO α such that $y(2) = 1$

(D) there is a unique α such that $y(1) = 2$

Question 5 (JAM 2023 • Q5 • ID: JAM_2023_Q5) MCQ +1 M Neg: −0.33

Q. 5 Let $p(x) = x^{57} + 3x^{10} - 21x^3 + x^2 + 21$ and

$$q(x) = p(x) + \sum_{j=1}^{57} p^{(j)}(x) \quad \text{for all } x \in \mathbb{R},$$

where $p^{(j)}(x)$ denotes the j^{th} derivative of $p(x)$. Then the function q admits

(A) NEITHER a global maximum NOR a global minimum on $\mathbb{R}$

(B) a global maximum but NOT a global minimum on $\mathbb{R}$

(C) a global minimum but NOT a global maximum on $\mathbb{R}$

(D) a global minimum and a global maximum on $\mathbb{R}$

Question 6 (JAM 2023 • Q6 • ID: JAM_2023_Q6) MCQ +1 M Neg: −0.33

Q. 6 The limit

$$\lim_{a\to 0}\left(\frac{\int_0^a \sin(x^2)\,dx}{\int_0^a (\ln(x+1))^2\,dx}\right)$$

is

(A) 0

(B) 1

(C) $\frac{\pi}{e}$

(D) non-existent

Question 7 (JAM 2023 • Q7 • ID: JAM_2023_Q7) MCQ +1 M Neg: −0.33

Q. 7 The value of

$$\int_0^1\int_0^{1-x}\cos(x^3+y^2)\,dy\,dx-\int_0^1\int_0^{1-y}\cos(x^3+y^2)\,dx\,dy$$

is

(A) 0

(B) $\frac{\cos(1)}{2}$

(C) $\frac{\sin(1)}{2}$

(D) $\cos\left(\frac{1}{2}\right)-\sin\left(\frac{1}{2}\right)$

Question 8 (JAM 2023 • Q8 • ID: JAM_2023_Q8) MCQ +1 M Neg: −0.33

Q. 8 Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $f(x, y) = (e^x \cos(y), e^x \sin(y))$. Then the number of points in $\mathbb{R}^2$ that do NOT lie in the range of f is

(A) 0

(B) 1

(C) 2

(D) infinite

Question 9 (JAM 2023 • Q9 • ID: JAM_2023_Q9) MCQ +1 M Neg: −0.33

Q. 9 Let $a_n = \left(1 + \frac{1}{n}\right)^n$ and $b_n = n \cos\left(\frac{n!\pi}{2^{10}}\right)$ for $n \in \mathbb{N}$. Then

(A) (a_n) is convergent and (b_n) is bounded

(B) (a_n) is NOT convergent and (b_n) is bounded

(C) (a_n) is convergent and (b_n) is unbounded

(D) (a_n) is NOT convergent and (b_n) is unbounded

Question 10 (JAM 2023 • Q10 • ID: JAM_2023_Q10) MCQ +1 M Neg: −0.33

Q. 10 Let (a_n) be a sequence of real numbers defined by

$$a_n = \begin{cases} 1 & \text{if } n \text{ is prime} \\ -1 & \text{if } n \text{ is not prime.} \end{cases}$$

Let $b_n = \dfrac{a_n}{n}$ for $n \in \mathbb{N}$. Then

(A) both (a_n) and (b_n) are convergent

(B) (a_n) is convergent but (b_n) is NOT convergent

(C) (a_n) is NOT convergent but (b_n) is convergent

(D) both (a_n) and (b_n) are NOT convergent

Question 11 (JAM 2023 • Q11 • ID: JAM_2023_Q11) MCQ +2 M Neg: −0.67

Q. 11 Let $a_n = \sin\left(\dfrac{1}{n^3}\right)$ and $b_n = \sin\left(\dfrac{1}{n}\right)$ for $n \in \mathbb{N}$. Then

(A) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent

(B) $\sum_{n=1}^{\infty} a_n$ is convergent but $\sum_{n=1}^{\infty} b_n$ is NOT convergent

(C) $\sum_{n=1}^{\infty} a_n$ is NOT convergent but $\sum_{n=1}^{\infty} b_n$ is convergent

(D) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are NOT convergent

Question 12 (JAM 2023 • Q12 • ID: JAM_2023_Q12) MCQ +2 M Neg: −0.67

Q. 12 Consider the following statements:

I. There exists a linear transformation from $\mathbb{R}^3$ to itself such that its range space and null space are the same.

II. There exists a linear transformation from $\mathbb{R}^2$ to itself such that its range space and null space are the same.

Then

(A) both I and II are TRUE

(B) I is TRUE but II is FALSE

(C) II is TRUE but I is FALSE

(D) both I and II are FALSE

Question 13 (JAM 2023 • Q13 • ID: JAM_2023_Q13) MCQ +2 M Neg: −0.67

Q. 13 Let

$$A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ -2 & 2 & 2 \end{pmatrix}$$

and $B = A^5 + A^4 + I_3$. Which of the following is NOT an eigenvalue of B?

(A) 1

(B) 2

(C) 49

(D) 3

Question 14 (JAM 2023 • Q14 • ID: JAM_2023_Q14) MCQ +2 M Neg: −0.67

Q. 14 The system of linear equations in x_1, x_2, x_3

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \\ 2 & 3 & \alpha \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ \beta \end{pmatrix}$$

where $\alpha, \beta \in \mathbb{R}$, has

(A) at least one solution for any α and β

(B) a unique solution for any β when $\alpha \neq 1$

(C) NO solution for any α when $\beta \neq 5$

(D) infinitely many solutions for any α when $\beta = 5$

Question 15 (JAM 2023 • Q15 • ID: JAM_2023_Q15) MCQ +2 M Neg: −0.67

Q. 15 Let S and T be non-empty subsets of $\mathbb{R}^2$, and W be a non-zero proper subspace of $\mathbb{R}^2$. Consider the following statements:

I. If $\text{span}(S) = \mathbb{R}^2$, then $\text{span}(S \cap W) = W$.

II. $\text{span}(S \cup T) = \text{span}(S) \cup \text{span}(T)$.

Then

(A) both I and II are TRUE

(B) I is TRUE but II is FALSE

(C) II is TRUE but I is FALSE

(D) both I and II are FALSE

Question 16 (JAM 2023 • Q16 • ID: JAM_2023_Q16) MCQ +2 M Neg: −0.67

Q. 16 Let $f(x,y) = e^{x^2+y^2}$ for $(x,y) \in \mathbb{R}^2$, and a_n be the determinant of the matrix

$$\begin{pmatrix} \dfrac{\partial^2 f}{\partial x^2} & \dfrac{\partial^2 f}{\partial x \partial y} \\ \dfrac{\partial^2 f}{\partial y \partial x} & \dfrac{\partial^2 f}{\partial y^2} \end{pmatrix}$$

evaluated at the point $(\cos(n), \sin(n))$. Then the limit $\lim_{n\to\infty} a_n$ is

(A) non-existent

(B) 0

(C) $6e^2$

(D) $12e^2$

Question 17 (JAM 2023 • Q17 • ID: JAM_2023_Q17) MCQ +2 M Neg: −0.67

Q. 17 Let $f(x,y) = \ln(1 + x^2 + y^2)$ for $(x,y) \in \mathbb{R}^2$. Define

$$P = \left.\frac{\partial^2 f}{\partial x^2}\right|_{(0,0)} \qquad Q = \left.\frac{\partial^2 f}{\partial x \partial y}\right|_{(0,0)}$$

$$R = \left.\frac{\partial^2 f}{\partial y \partial x}\right|_{(0,0)} \qquad S = \left.\frac{\partial^2 f}{\partial y^2}\right|_{(0,0)}$$

Then

(A) $PS - QR > 0$ and $P < 0$

(B) $PS - QR > 0$ and $P > 0$

(C) $PS - QR < 0$ and $P > 0$

(D) $PS - QR < 0$ and $P < 0$

Question 18 (JAM 2023 • Q18 • ID: JAM_2023_Q18) MCQ +2 M Neg: −0.67

Q. 18 The area of the curved surface

$$S = \{(x, y, z) \in \mathbb{R}^3 : z^2 = (x-1)^2 + (y-2)^2\}$$

lying between the planes $z = 2$ and $z = 3$ is

(A) $4\pi\sqrt{2}$

(B) $5\pi\sqrt{2}$

(C) 9π

(D) $9\pi\sqrt{2}$

Question 19 (JAM 2023 • Q19 • ID: JAM_2023_Q19) MCQ +2 M Neg: −0.67

Q. 19 Let $a_n = \dfrac{1 + 2^{-2} + \cdots + n^{-2}}{n}$ for $n \in \mathbb{N}$. Then

(A) both the sequence (a_n) and the series $\sum_{n=1}^{\infty} a_n$ are convergent

(B) the sequence (a_n) is convergent but the series $\sum_{n=1}^{\infty} a_n$ is NOT convergent

(C) both the sequence (a_n) and the series $\sum_{n=1}^{\infty} a_n$ are NOT convergent

(D) the sequence (a_n) is NOT convergent but the series $\sum_{n=1}^{\infty} a_n$ is convergent

Question 20 (JAM 2023 • Q20 • ID: JAM_2023_Q20) MCQ +2 M Neg: −0.67

Q. 20 Let (a_n) be a sequence of real numbers such that the series $\sum_{n=0}^{\infty} a_n(x-2)^n$ converges at $x = -5$. Then this series also converges at

(A) $x = 9$

(B) $x = 12$

(C) $x = 5$

(D) $x = -6$

Question 21 (JAM 2023 • Q21 • ID: JAM_2023_Q21) MCQ +2 M Neg: −0.67

Q. 21 Let (a_n) and (b_n) be sequences of real numbers such that

$$|a_n - a_{n+1}| = \frac{1}{2^n} \quad \text{and} \quad |b_n - b_{n+1}| = \frac{1}{\sqrt{n}} \quad \text{for } n \in \mathbb{N}.$$

Then

(A) both (a_n) and (b_n) are Cauchy sequences

(B) (a_n) is a Cauchy sequence but (b_n) need NOT be a Cauchy sequence

(C) (a_n) need NOT be a Cauchy sequence but (b_n) is a Cauchy sequence

(D) both (a_n) and (b_n) need NOT be Cauchy sequences

Question 22 (JAM 2023 • Q22 • ID: JAM_2023_Q22) MCQ +2 M Neg: −0.67

Q. 22 Consider the family of curves $x^2 + y^2 = 2x + 4y + k$ with a real parameter $k > -5$. Then the orthogonal trajectory to this family of curves passing through $(2, 3)$ also passes through

(A) $(3, 4)$

(B) $(-1, 1)$

(C) $(1, 0)$

(D) $(3, 5)$

Question 23 (JAM 2023 • Q23 • ID: JAM_2023_Q23) MCQ +2 M Neg: −0.67

Q. 23 Consider the following statements:

I. Every infinite group has infinitely many subgroups.

II. There are only finitely many non-isomorphic groups of a given finite order.

Then

(A) both I and II are TRUE

(B) I is TRUE but II is FALSE

(C) I is FALSE but II is TRUE

(D) both I and II are FALSE

Question 24 (JAM 2023 • Q24 • ID: JAM_2023_Q24) MCQ +2 M Neg: −0.67

Q. 24 Suppose $f : (-1, 1) \to \mathbb{R}$ is an infinitely differentiable function such that the series $\sum_{j=0}^{\infty} a_j \frac{x^j}{j!}$ converges to $f(x)$ for each $x \in (-1, 1)$, where,

$$a_j = \int_0^{\pi/2} \theta^j \cos^j(\tan\theta) d\theta + \int_{\pi/2}^{\pi} (\theta - \pi)^j \cos^j(\tan\theta) d\theta$$

for $j \geq 0$. Then

(A) $f(x) = 0$ for all $x \in (-1, 1)$

(B) f is a non-constant even function on $(-1, 1)$

(C) f is a non-constant odd function on $(-1, 1)$

(D) f is NEITHER an odd function NOR an even function on $(-1, 1)$

Question 25 (JAM 2023 • Q25 • ID: JAM_2023_Q25) MCQ +2 M Neg: −0.67

Q. 25 Let $f(x) = \cos(x)$ and $g(x) = 1 - \frac{x^2}{2}$ for $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then

(A) $f(x) \geq g(x)$ for all $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(B) $f(x) \leq g(x)$ for all $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(C) $f(x) - g(x)$ changes sign exactly once on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(D) $f(x) - g(x)$ changes sign more than once on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Question 26 (JAM 2023 • Q26 • ID: JAM_2023_Q26) MCQ +2 M Neg: −0.67

Q. 26 Let

$$f(x,y) = \iint_{(u-x)^2+(v-y)^2 \le 1} e^{-\sqrt{(u-x)^2+(v-y)^2}}\, du\, dv.$$

Then $\lim_{n\to\infty} f(n, n^2)$ is

(A) non-existent

(B) 0

(C) $\pi(1 - e^{-1})$

(D) $2\pi(1 - 2e^{-1})$

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Question 27 (JAM 2023 • Q27 • ID: JAM_2023_Q27) MCQ +2 M Neg: −0.67

Q. 27 How many group homomorphisms are there from $\mathbb{Z}_2$ to S_5?

(A) 40

(B) 41

(C) 26

(D) 25

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Question 28 (JAM 2023 • Q28 • ID: JAM_2023_Q28) MCQ +2 M Neg: −0.67

Q. 28 Let $y : \mathbb{R} \to \mathbb{R}$ be a twice differentiable function such that y'' is continuous on $[0, 1]$ and $y(0) = y(1) = 0$. Suppose $y''(x) + x^2 < 0$ for all $x \in [0, 1]$. Then

(A) $y(x) > 0$ for all $x \in (0, 1)$

(B) $y(x) < 0$ for all $x \in (0, 1)$

(C) $y(x) = 0$ has exactly one solution in $(0, 1)$

(D) $y(x) = 0$ has more than one solution in $(0, 1)$

Question 29 (JAM 2023 • Q29 • ID: JAM_2023_Q29) MCQ +2 M Neg: −0.67

Q. 29 From the additive group $\mathbb{Q}$ to which one of the following groups does there exist a non-trivial group homomorphism?

(A) $\mathbb{R}^\times$, the multiplicative group of non-zero real numbers

(B) $\mathbb{Z}$, the additive group of integers

(C) $\mathbb{Z}_2$, the additive group of integers modulo 2

(D) $\mathbb{Q}^\times$, the multiplicative group of non-zero rational numbers

Question 30 (JAM 2023 • Q30 • ID: JAM_2023_Q30) MCQ +2 M Neg: −0.67

Q. 30 Let $f : \mathbb{R} \to \mathbb{R}$ be an infinitely differentiable function such that f'' has exactly two distinct zeroes. Then

(A) f' has at most 3 distinct zeroes

(B) f' has at least 1 zero

(C) f has at most 3 distinct zeroes

(D) f has at least 2 distinct zeroes

Question 31 (JAM 2023 • Q31 • ID: JAM_2023_Q31) MSQ +2 M Neg: Nil (0)

Q. 31 For each $t \in (0, 1)$, the surface P_t in $\mathbb{R}^3$ is defined by

$$P_t = \{(x, y, z) : (x^2 + y^2)z = 1,\ t^2 \le x^2 + y^2 \le 1\}.$$

Let $a_t \in \mathbb{R}$ be the surface area of P_t. Then

(A) $a_t = \displaystyle\iint_{t^2 \le x^2+y^2 \le 1} \sqrt{1 + \frac{4x^2}{(x^2+y^2)^4} + \frac{4y^2}{(x^2+y^2)^4}}\, dx\, dy$

(B) $a_t = \displaystyle\iint_{t^2 \le x^2+y^2 \le 1} \sqrt{1 + \frac{4x^2}{(x^2+y^2)^2} + \frac{4y^2}{(x^2+y^2)^2}}\, dx\, dy$

(C) the limit $\lim_{t \to 0^+} a_t$ does NOT exist

(D) the limit $\lim_{t \to 0^+} a_t$ exists

Question 32 (JAM 2023 • Q32 • ID: JAM_2023_Q32) MSQ +2 M Neg: Nil (0)

Q. 32 Let $A \subseteq \mathbb{Z}$ with $0 \in A$. For $r, s \in \mathbb{Z}$, define

$$rA = \{ra : a \in A\}, \qquad rA + sA = \{ra + sb : \ a, b \in A\}.$$

Which of the following conditions imply that A is a subgroup of the additive group $\mathbb{Z}$?

(A) $-2A \subseteq A, \ \ A + A = A$

(B) $A = -A, \ \ A + 2A = A$

(C) $A = -A, \ \ A + A = A$

(D) $2A \subseteq A, \ \ A + A = A$

Question 33 (JAM 2023 • Q33 • ID: JAM_2023_Q33) MSQ +2 M Neg: Nil (0)

Q. 33 Let $y : (\sqrt{2/3}, \infty) \to \mathbb{R}$ be the solution of

$$(2x - y)y' + (2y - x) = 0,$$
$$y(1) = 3.$$

Then

(A) $y(3) = 1$

(B) $y(2) = 4 + \sqrt{10}$

(C) y' is bounded on $(\sqrt{2/3}, 1)$

(D) y' is bounded on $(1, \infty)$

Question 34 (JAM 2023 • Q34 • ID: JAM_2023_Q34) MSQ +2 M Neg: Nil (0)

Q. 34 Let $f : (-1, 1) \to \mathbb{R}$ be a differentiable function satisfying $f(0) = 0$. Suppose there exists an $M > 0$ such that $|f'(x)| \leq M|x|$ for all $x \in (-1, 1)$. Then

(A) f' is continuous at $x = 0$

(B) f' is differentiable at $x = 0$

(C) ff' is differentiable at $x = 0$

(D) $(f')^2$ is differentiable at $x = 0$

Question 35 (JAM 2023 • Q35 • ID: JAM_2023_Q35) MSQ +2 M Neg: Nil (0)

Q. 35 Which of the following functions is/are Riemann integrable on $[0, 1]$?

(A) $f(x) = \displaystyle\int_0^x \left|\frac{1}{2} - t\right| dt$

(B) $f(x) = \begin{cases} x \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

(C) $f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \cap [0, 1] \\ -1 & \text{otherwise} \end{cases}$

(D) $f(x) = \begin{cases} x & \text{if } x \in [0, 1) \\ 0 & \text{if } x = 1 \end{cases}$

Question 36 (JAM 2023 • Q36 • ID: JAM_2023_Q36) MSQ +2 M Neg: Nil (0)

Q. 36 A subset $S \subseteq \mathbb{R}^2$ is said to be *bounded* if there is an $M > 0$ such that $|x| \leq M$ and $|y| \leq M$ for all $(x, y) \in S$. Which of the following subsets of $\mathbb{R}^2$ is/are bounded?

(A) $\{(x, y) \in \mathbb{R}^2 : e^{x^2} + y^2 \leq 4\}$

(B) $\{(x, y) \in \mathbb{R}^2 : x^4 + y^2 \leq 4\}$

(C) $\{(x, y) \in \mathbb{R}^2 : |x| + |y| \leq 4\}$

(D) $\{(x, y) \in \mathbb{R}^2 : e^{x^3} + y^2 \leq 4\}$

Question 37 (JAM 2023 • Q37 • ID: JAM_2023_Q37) MSQ +2 M Neg: Nil (0)

Q. 37 Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined as follows:

$$f(x, y) = \begin{cases} \dfrac{x^4 y^3}{x^6 + y^6} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Then

(A) $\lim_{t \to 0} \dfrac{f(t, t) - f(0, 0)}{t}$ exists and equals $\dfrac{1}{2}$

(B) $\left.\dfrac{\partial f}{\partial x}\right|_{(0,0)}$ exists and equals 0

(C) $\left.\dfrac{\partial f}{\partial y}\right|_{(0,0)}$ exists and equals 0

(D) $\lim_{t \to 0} \dfrac{f(t, 2t) - f(0, 0)}{t}$ exists and equals $\dfrac{1}{3}$

Question 38 (JAM 2023 • Q38 • ID: JAM_2023_Q38) MSQ +2 M Neg: Nil (0)

Q. 38 Which of the following is/are true?

(A) Every linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps lines onto points or lines

(B) Every surjective linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps lines onto lines

(C) Every bijective linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps pairs of parallel lines to pairs of parallel lines

(D) Every bijective linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps pairs of perpendicular lines to pairs of perpendicular lines

Question 39 (JAM 2023 • Q39 • ID: JAM_2023_Q39) MSQ +2 M Neg: Nil (0)

Q. 39 Which of the following is/are linear transformations?

(A) $T : \mathbb{R} \to \mathbb{R}$ given by $T(x) = \sin(x)$

(B) $T : M_2(\mathbb{R}) \to \mathbb{R}$ given by $T(A) = \text{trace}(A)$

(C) $T : \mathbb{R}^2 \to \mathbb{R}$ given by $T(x, y) = x + y + 1$

(D) $T : P_2(\mathbb{R}) \to \mathbb{R}$ given by $T(p(x)) = p(1)$

Question 40 (JAM 2023 • Q40 • ID: JAM_2023_Q40) MSQ +2 M Neg: Nil (0)

Q. 40 Let R_1 and R_2 be the radii of convergence of the power series $\sum_{n=1}^{\infty}(-1)^n x^{n-1}$ and $\sum_{n=1}^{\infty}(-1)^n \frac{x^{n+1}}{n(n+1)}$, respectively. Then

(A) $R_1 = R_2$

(B) $R_2 > 1$

(C) $\sum_{n=1}^{\infty}(-1)^n x^{n-1}$ converges for all $x \in [-1, 1]$

(D) $\sum_{n=1}^{\infty}(-1)^n \frac{x^{n+1}}{n(n+1)}$ converges for all $x \in [-1, 1]$

Question 41 (JAM 2023 • Q41 • ID: JAM_2023_Q41) NAT +1 M Neg: Nil (0)

Q. 41 Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function defined as follows:

$$f(x,y) = \begin{cases} (x^2-1)^2 \cos^2\left(\frac{y^2}{(x^2-1)^2}\right) & \text{if } x \neq \pm 1 \\ 0 & \text{if } x = \pm 1. \end{cases}$$

The number of points of discontinuity of $f(x, y)$ is equal to _____.

Question 42 (JAM 2023 • Q42 • ID: JAM_2023_Q42) NAT +1 M Neg: Nil (0)

Q. 42 Let $T : P_2(\mathbb{R}) \to P_4(\mathbb{R})$ be the linear transformation given by $T(p(x)) = p(x^2)$. Then the rank of T is equal to _____.

Question 43 (JAM 2023 • Q43 • ID: JAM_2023_Q43) NAT +1 M Neg: Nil (0)

Q. 43 If y is the solution of

$$y'' - 2y' + y = e^x,$$
$$y(0) = 0, \quad y'(0) = -1/2,$$

then $y(1)$ is equal to _____. (rounded off to two decimal places)

Question 44 (JAM 2023 • Q44 • ID: JAM_2023_Q44) NAT +1 M Neg: Nil (0)

Q. 44 The value of

$$\lim_{n\to\infty} \left(n \int_0^1 \frac{x^n}{x+1}\, dx \right)$$

is equal to _____. (rounded off to two decimal places)

Question 45 (JAM 2023 • Q45 • ID: JAM_2023_Q45) NAT +1 M Neg: Nil (0)

Q. 45 For $\sigma \in S_8$, let $o(\sigma)$ denote the order of σ. Then $\max\{o(\sigma) : \sigma \in S_8\}$ is equal to _____.

Question 46 (JAM 2023 • Q46 • ID: JAM_2023_Q46) NAT +1 M Neg: Nil (0)

Q. 46 For $g \in \mathbb{Z}$, let $\bar{g} \in \mathbb{Z}_8$ denote the residue class of g modulo 8. Consider the group $\mathbb{Z}_8^\times = \{\bar{x} \in \mathbb{Z}_8 : 1 \le x \le 7, \gcd(x, 8) = 1\}$ with respect to multiplication modulo 8. The number of group isomorphisms from $\mathbb{Z}_8^\times$ onto itself is equal to _____.

Question 47 (JAM 2023 • Q47 • ID: JAM_2023_Q47) NAT +1 M Neg: Nil (0)

Q. 47 Let $f(x) = \sqrt[3]{x}$ for $x \in (0, \infty)$, and $\theta(h)$ be a function such that

$$f(3+h) - f(3) = hf'(3+\theta(h)h)$$

for all $h \in (-1, 1)$. Then $\lim_{h\to 0} \theta(h)$ is equal to _____.
(rounded off to two decimal places)

Question 48 (JAM 2023 • Q48 • ID: JAM_2023_Q48) NAT +1 M Neg: Nil (0)

Q. 48 Let V be the volume of the region $S \subseteq \mathbb{R}^3$ defined by

$$S = \{(x, y, z) \in \mathbb{R}^3 : \ xy \le z \le 4, \ 0 \le x^2 + y^2 \le 1\}.$$

Then $\dfrac{V}{\pi}$ is equal to _____. (rounded off to two decimal places)

Question 49 (JAM 2023 • Q49 • ID: JAM_2023_Q49) NAT +1 M Neg: Nil (0)

Q. 49 The sum of the series $\displaystyle\sum_{n=1}^{\infty} \frac{2n+1}{(n^2+1)(n^2+2n+2)}$ is equal to _____.
(rounded off to two decimal places)

Question 50 (JAM 2023 • Q50 • ID: JAM_2023_Q50) NAT +1 M Neg: Nil (0)

Q. 50 The value of $\displaystyle\lim_{n\to\infty} \left(1 + \frac{1}{2^n} + \frac{1}{3^n} + \cdots + \frac{1}{(2023)^n}\right)^{\frac{1}{n}}$ is equal to _____.
(rounded off to two decimal places)

Question 51 (JAM 2023 • Q51 • ID: JAM_2023_Q51) NAT +2 M Neg: Nil (0)

Q. 51 Let $f : \mathbb{R}^3 \to \mathbb{R}$ be defined as $f(x, y, z) = x^3 + y^3 + z^3$, and let $L : \mathbb{R}^3 \to \mathbb{R}$ be the linear map satisfying

$$\lim_{(x,y,z)\to(0,0,0)} \frac{f(1+x, 1+y, 1+z) - f(1,1,1) - L(x,y,z)}{\sqrt{x^2+y^2+z^2}} = 0.$$

Then $L(1, 2, 4)$ is equal to _____. (rounded off to two decimal places)

Question 52 (JAM 2023 • Q52 • ID: JAM_2023_Q52) NAT +2 M Neg: Nil (0)

Q. 52 The global minimum value of

$$f(x) = |x - 1| + |x - 2|^2$$

on $\mathbb{R}$ is equal to _____. (rounded off to two decimal places)

Question 53 (JAM 2023 • Q53 • ID: JAM_2023_Q53) NAT +2 M Neg: Nil (0)

Q. 53 Let $y : (1, \infty) \to \mathbb{R}$ be the solution of the differential equation

$$y'' - \frac{2y}{(1-x)^2} = 0$$

satisfying $y(2) = 1$ and $\lim_{x\to\infty} y(x) = 0$. Then $y(3)$ is equal to _____. (rounded off to two decimal places)

Question 54 (JAM 2023 • Q54 • ID: JAM_2023_Q54) NAT +2 M Neg: Nil (0)

Q. 54 The number of permutations in S_4 that have exactly two cycles in their cycle decompositions is equal to _____.

Question 55 (JAM 2023 • Q55 • ID: JAM_2023_Q55) NAT +2 M Neg: Nil (0)

Q. 55 Let S be the triangular region whose vertices are $(0,0)$, $\left(0, \frac{\pi}{2}\right)$, and $\left(\frac{\pi}{2}, 0\right)$. The value of $\iint_S \sin(x)\cos(y)\,dx\,dy$ is equal to _____.
(rounded off to two decimal places)

Question 56 (JAM 2023 • Q56 • ID: JAM_2023_Q56) NAT +2 M Neg: Nil (0)

Q. 56 Let

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & 4 & 4 & 4 \end{pmatrix}$$

and B be a 5×5 real matrix such that AB is the zero matrix. Then the maximum possible rank of B is equal to _____.

Question 57 (JAM 2023 • Q57 • ID: JAM_2023_Q57) NAT +2 M Neg: Nil (0)

Q. 57 Let W be the subspace of $M_3(\mathbb{R})$ consisting of all matrices with the property that the sum of the entries in each row is zero and the sum of the entries in each column is zero. Then the dimension of W is equal to _____.

Question 58 (JAM 2023 • Q58 • ID: JAM_2023_Q58) NAT +2 M Neg: Nil (0)

Q. 58 The maximum number of linearly independent eigenvectors of the matrix

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 1 & 3 \end{pmatrix}$$

is equal to _____.

Question 59 (JAM 2023 • Q59 • ID: JAM_2023_Q59) NAT +2 M Neg: Nil (0)

Q. 59 Let S be the set of all real numbers α such that the solution y of the initial value problem

$$\frac{dy}{dx} = y(2-y),$$
$$y(0) = \alpha,$$

exists on $[0, \infty)$. Then the minimum of the set S is equal to _____.

(rounded off to two decimal places)

Question 60 (JAM 2023 • Q60 • ID: JAM_2023_Q60) NAT +2 M Neg: Nil (0)

Q. 60 Let $f : \mathbb{R} \to \mathbb{R}$ be a bijective function such that for all $x \in \mathbb{R}$, $f(x) = \sum_{n=1}^{\infty} a_n x^n$ and $f^{-1}(x) = \sum_{n=1}^{\infty} b_n x^n$, where f^{-1} is the inverse function of f. If $a_1 = 2$ and $a_2 = 4$, then b_1 is equal to _____.

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

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Q. No.	Type	Marks	Official Key
1	MCQ	+1	**B**
2	MCQ	+1	**D**
3	MCQ	+1	**A**
4	MCQ	+1	**D**
5	MCQ	+1	**A**
6	MCQ	+1	**B**
7	MCQ	+1	**A**
8	MCQ	+1	**B**
9	MCQ	+1	**C**
10	MCQ	+1	**C**
11	MCQ	+2	**B**
12	MCQ	+2	**C**
13	MCQ	+2	**B**
14	MCQ	+2	**B**
15	MCQ	+2	**D**
16	MCQ	+2	**D**
17	MCQ	+2	**B**
18	MCQ	+2	**B**
19	MCQ	+2	**B**
20	MCQ	+2	**C**

Q. No.	Type	Marks	Official Key
21	MCQ	+2	**B**
22	MCQ	+2	**A**
23	MCQ	+2	**A**
24	MCQ	+2	**B**
25	MCQ	+2	**A**
26	MCQ	+2	**D**
27	MCQ	+2	**C**
28	MCQ	+2	**A**
29	MCQ	+2	**A**
30	MCQ	+2	**A**
31	MSQ	+2	**A, C**
32	MSQ	+2	**A, B, C**
33	MSQ	+2	**B, D**
34	MSQ	+2	**A, C, D**
35	MSQ	+2	**A, B, D**
36	MSQ	+2	**A, B, C**
37	MSQ	+2	**A, B, C**
38	MSQ	+2	**A, B, C**
39	MSQ	+2	**B, D**
40	MSQ	+2	**A, D**

Q. No.	Type	Marks	Official Key
41	NAT	+1	**0 to 0**
42	NAT	+1	**3 to 3**
43	NAT	+1	**−0.01 to 0.01**
44	NAT	+1	**0.49 to 0.51**
45	NAT	+1	**15 to 15**
46	NAT	+1	**6 to 6**
47	NAT	+1	**0.49 to 0.51**
48	NAT	+1	**3.99 to 4.01**
49	NAT	+1	**0.49 to 0.51**
50	NAT	+1	**0.99 to 1.01**
51	NAT	+2	**20.99 to 21.01**
52	NAT	+2	**0.74 to 0.76**
53	NAT	+2	**0.49 to 0.51**
54	NAT	+2	**11 to 11**
55	NAT	+2	**0.49 to 0.51**
56	NAT	+2	**2 to 2**
57	NAT	+2	**4 to 4**
58	NAT	+2	**3 to 3**
59	NAT	+2	**−0.01 to 0.01**
60	NAT	+2	**0.49 to 0.51**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, −2/3 for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.